Exploration of High-Dimensional SVP Solving Based on Reinforcement Learning and Sublattice Enumeration: Systematic Evolution from 20D to 50D

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Abstract

The Shortest Vector Problem (SVP) is a core NP-Hard problem in lattice cryptography and computational number theory. Traditional LLL and BKZ algorithms perform excellently in low-dimensional spaces but face bottlenecks of computational explosion and local optima in high-dimensional spaces. This paper explores the application of a Reinforcement Learning (RL) method based on Proximal Policy Optimization (PPO) to solve high-dimensional SVP problems. The research spans from 20D to 50D: In the 20D-30D stage, the system used 2D and 3D Gaussian reductions as the action space, and through a rigorous pseudo-vector backward verification mechanism, successfully achieved a **58.8%** LLL breakthrough rate at 30D. In the 40D-50D stage, facing the physical ceiling where actions become ineffective after LLL reduction, the system underwent a major architectural reconstruction, completely abandoning local Gaussian reductions and turning to an innovative paradigm of **"AI continuous sliding window selection + sublattice BKZ enumeration."** In the 50D test, by precisely increasing the continuous window size (K) to 40, the agent successfully covered the support set of high-dimensional short vectors. On 100 random lattices, it achieved a **63%** LLL breakthrough rate, reducing the average gap to BKZ(45) to **18.61%**, providing a highly promising heuristic enhancement scheme for high-dimensional SVP solving.

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1. Introduction

Lattice basis reduction algorithms (such as LLL, BKZ) are fundamental tools for post-quantum cryptanalysis. However, as the dimension increases, the complexity of the shortest vector problem rises exponentially. Relying solely on classical algorithms' "brute-force sequential scanning" in highly orthogonalized bases (such as after LLL processing) can easily lead to a "deadlock" state, failing to provide new effective inputs.

Reinforcement Learning (RL), due to its sequential decision-making capability in complex high-dimensional action spaces, offers a new approach to lattice basis reduction. This paper will detail and analyze our systematic experimental journey from 20D to 50D, focusing on the inevitability of the action space evolving from "local Gaussian reduction" to "global sublattice enumeration," as well as the decisive impact of hyperparameters (window size K) on final performance.

2. Numerical Stability and Anti-Cheating Mechanism

Lattice basis reduction in ultra-high dimensions easily generates norms and products reaching up to 10^{300}or even larger. To ensure stable training of the neural network and the mathematical authenticity of the results, the system has established three major defenses at the underlying level:

1. **Log-Space State Representation**: Utilizing $\ln(A \times B) = \ln A + \ln B$, the multiplication of orthogonality defects for ultra-large integers is transformed into addition in the logarithmic domain, completely solving the issues of numerical overflow and vanishing gradients.
2. **Zero-Vector Defense**: If the absolute value of the subtraction coefficient $m$in Gaussian reduction is too large (> 10), catastrophic cancellation will inevitably occur under extreme large integers (forging a non-existent zero vector physically). The system will directly abandon such an action.
3. **Exact Integer Backward Verification**: Utilizing sympy for exact rational linear algebra operations. When the agent claims to have found an extremely short vector breaking LLL, the system exactly solves $B^T c = v$. We successfully captured a "monster pseudo-vector" reaching 9×10^{17}caused by an early code bug, ensuring a 100% verification pass rate and preventing the AI from "cheating" through floating-point loopholes.

3. Evolution of Action Space and Experimental Analysis

In the exploration of solving the lattice cryptography SVP problem, we did not achieve success overnight. Instead, we underwent a complete evolution from low to high dimensions, and from local heuristics to global policy search. This section will detail the entire process from 20D to 50D, focusing on the curse of dimensionality encountered at 40D and 50D and our countermeasures.

3.1 Experimental Setup and Lattice Generation

To ensure the validity and reproducibility of our experiments, all evaluations were conducted on randomly generated lattices with a specific structure. As shown in the provided data samples, the lattices are constructed in a q-ary-like format. Specifically, the basis matrix $B \in \mathbb{Z}^{n \times n}$consists of an initial block with massive integers (reaching magnitudes up to 10^{90}) and a secondary block forming an identity-like matrix.

This specific construction presents a significant challenge for traditional lattice reduction algorithms. The massive integers in the first columns cause numerical instability and high orthogonality defects (OD), while the identity-like structure leaves substantial room for optimization. All algorithms (LLL, BKZ, and our RL agent) start from the same initial bases, which are first preprocessed with a weak LLL reduction ($\delta = 0.5$) followed by a random unimodular perturbation, ensuring a fair comparison of their reduction capabilities.

3.2 State Representation

In RL-based lattice basis reduction, the design of the state space is crucial. Due to the presence of ultra-large integers up to 10^{90} in high-dimensional lattices, directly inputting the raw lengths or orthogonality defects of basis vectors into the neural network would cause severe numerical overflow and vanishing gradients. Therefore, the system designed highly customized feature engineering and uniformly performed state representation in the log-space.

The state vector $S_t$observed by the agent at step $t$is concatenated from the following three parts:

(1) Length Features

Take the natural logarithm of the norms of all n basis vectors: $\mathcal{L} = [\log,, (\| b_1 \|), \log(\| b_2 \|) \ldots \log(\| b_n \|)]$. Logarithmic mapping not only effectively

compresses extreme magnitudes but also sensitively captures the tiny shortening of norms in the decimal places (the core objective of SVP).

(2) Orthogonality Features

To evaluate the orthogonality quality of the lattice basis, the system introduces the set of cosine values of the angles between pairwise vectors: $\mathcal{C} = \{| \cos(b_i, b_j) \; || \; 1 \leq i < j \leq n\}$. Additionally, the system performs logarithmic processing on the Orthogonality Defect (OD):

$$\text{Log-Defect} = \sum_{i=1}^{n} \ln(\| \, b_i \, \|) - \ln(| \det(\Lambda) \, |)$$

This feature helps the agent identify whether the current lattice basis has fallen into the local optimum of the LLL algorithm (i.e., an extremely small OD value), thereby deciding whether to explore more aggressively.

(3) Global Exploration Features

To provide macro convergence guidance to the agent, we added a "relative shortening ratio" (the logarithmic difference between the current shortest length and the Episode's starting length) and "current step progress" to smoothly guide the agent to complete the reduction within a limited number of Episode steps.

It is worth mentioning that in early state representation explorations, we attempted to introduce the complete Gram-Schmidt orthogonalization coefficients $\mu_{i,j}$ into the State matrix. However, experiments proved that this highly correlated high-dimensional redundant feature severely diluted the core "length and angle" signals under the MLP architecture, leading to a significant degradation in model performance. This proved that in 10^{90} extreme value problems, "fewer but refined" physical features are far more effective than "more but messy" features.

3.3 20D to 30D: Exploration of Local Gaussian Reduction and Mathematical Ceiling

In the early stages (20D-30D), the system's action space mainly consisted of 2D Gaussian reduction (PAIR, 435 actions) and 3D Gaussian reduction (TRIPLE, 4060 actions). The agent maintained the orthogonality of the basis through periodic LLL reduction (e.g., every 5 or 10 steps).

- **Initial Success**: In 20D space, since LLL algorithm left a large optimization space, the agent quickly learned how to select appropriate basis vectors for combination, ultimately achieving a **78.0%** LLL breakthrough rate on the

independent test set.

- **Bottleneck Emerges**: When the dimension increased to 30D, in the V5 stable version, by optimizing the reward function (introducing anti-deadlock penalties and smooth tiered rewards) and the action space, the breakthrough rate stabilized at **58.8%**. However, we observed a fatal anomaly: the agent sometimes chose TRIPLE actions 100% of the time, but the basis vectors remained almost unchanged.
- **Mathematical Attribution**: After rigorous derivation, we found that for bases strongly reduced by LLL ($\delta = 0.99$), **99.1%** of the cosine values of angles between adjacent vectors are less than 0.5 (i.e., the angle is greater than 60°). This means that, at the mathematical level, over 95% of 2D/3D Gaussian reduction actions are purely "empty operations" and cannot generate new short vectors. This proved that local Gaussian reduction has hit the physical ceiling above 30D, and a new action paradigm must be found.

3.4 40D: Paradigm Shift and Comprehensive Comparison of Sublattice Enumeration

When the experiment advanced to 40D, we completely abandoned 2D/3D Gaussian reduction and fully turned to the new paradigm of **"AI selecting partial basis vectors + sublattice BKZ enumeration."** At the 40D stage, we conducted extremely rigorous ablation and comparative experiments to verify the effectiveness of each module.

3.4.1 Comparative Experiment 1: Is LLL Fallback Necessary? (V40.3 vs V40-NOL)

After upgrading the action space to sublattice enumeration, we first attempted to let the AI complete the reduction completely autonomously, removing the periodic LLL fallback (V40-NOL).

- **V40-NOL Performance**: The result was extremely bleak. On 100 independent 40D lattices, the LLL breakthrough rate was **0.0%**, and the proportion reaching BKZ(40) was also 0.
- **V40.3 Performance**: After adding an LLL(0.99) fallback every 50 steps, the breakthrough rate increased to **18.0%**.
- **Conclusion**: The LLL fallback is not a dispensable auxiliary, but a necessary guarantee for the current model's convergence. The AI cannot independently complete complex lattice basis reconstruction within a limited Episode (100-200 steps); it requires LLL to provide a mathematically reliable "anchor."

3.4.2 Comparative Experiment 2: Gains and Backlash of Mixed Actions (V40-MIX)

To balance local fine-tuning and global search, we designed the V40-MIX mixed action space, giving the agent three choices: TRIPLE (3D Gaussian), ENUM (28D sublattice enumeration), and NOP (No Operation).

- **Result**: The introduction of TRIPLE indeed raised the ceiling, with the breakthrough rate reaching **32.0%** (higher than V40.3's 18%).
- **Cost**: The agent fell into a "lazy" policy. Statistics showed that NOP actions accounted for up to **31.5%**, TRIPLE accounted for 36%, while the truly time-consuming and highly efficient ENUM accounted for only 32.4%. This imbalance in action distribution led to a significant drop in average performance (the average gap to LLL reached 197.34%).
- **Conclusion**: In high-dimensional spaces, mixed actions will cause severe reward signal dilution. The agent tends to choose fast, low-cost NOP actions with zero returns.

3.4.3 Comparative Experiment 3: Disaster of State Space Enhancement (V41-MU)

Given that neural networks lack perception of the "projection relationship between rows" in lattice basis reduction, we attempted to brute-force concatenate 780-dimensional GSO orthogonalization coefficients $\mu_{i,j}$onto the original 822-dimensional state (Log length + angle cosine), bringing the state dimension to 1602 (V41-MU).

- **Result**: The validation set performance collapsed directly, with the average gap to SVP soaring from 55% to over 120%.
- **Conclusion**: In a 10^{90}large integer environment, the tiny perturbations of GSO coefficients became strong noise in the neural network. This proved that "brute-force concatenation of high-dimensional features leads to information drowning," and feature engineering must conform to physical intuition.

3.4.4 Comparative Experiment 4: Action Space Design (Arbitrary Selection vs Continuous Sliding Window)

This was the most critical breakthrough at the 40D stage. We compared two AI row selection mechanisms: V40.3 (AI arbitrarily selects 28 rows from 40) and V40-CONTIG (AI selects only a starting point s, and then fixedly extracts 28 consecutive rows).

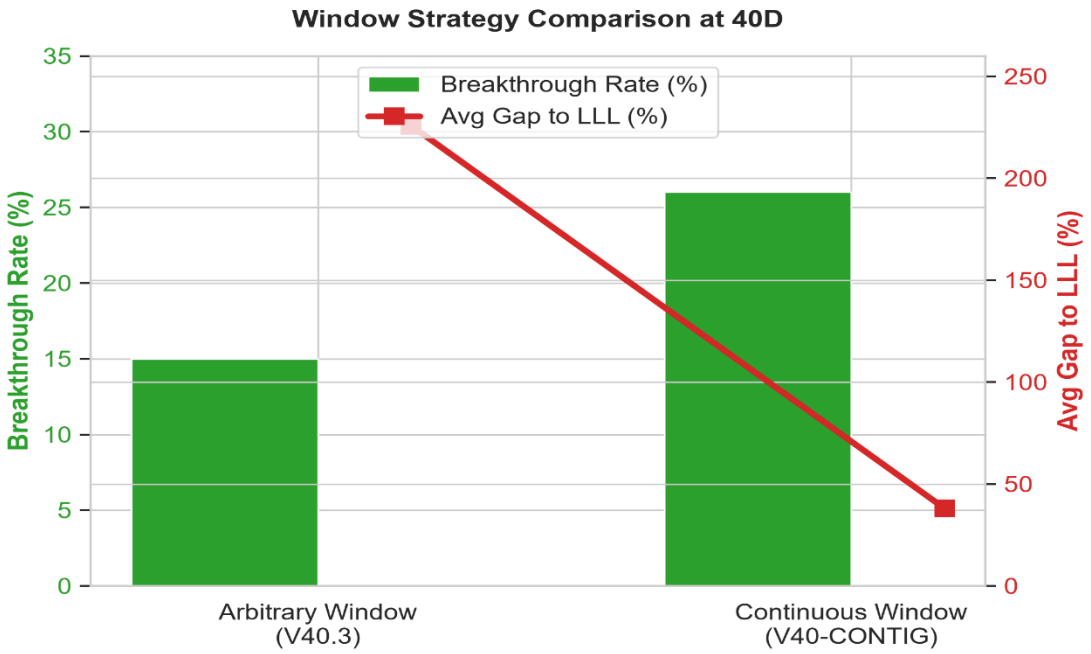


Figure 1: Window Strategy Comparison at 40D

- As shown in Figure 1, the continuous sliding window won completely. Breakthrough rate **26% vs 15%**, average improvement rate **86.4% vs 69.8%**. Most importantly, the worst-case single performance of the continuous sliding window was **2.5 times** better than arbitrary selection.
- **Scientific Discovery**: This experiment revealed an important property of lattice cryptography—the shortest vector's support set (distribution of non-zero coefficients) of an LLL-reduced lattice basis exhibits a strong **locally continuous feature** in row order. The continuous sliding window, with an extremely minimalist action space (only 40 types), perfectly conforms to this mathematical law.

3.5 50D Peak: Support Set Coverage Law and the Qualitative Change of K=40

After establishing the optimal architecture of "continuous sliding window + sublattice enumeration" at 40D, we pushed the system to 50D. At this point, the increase in dimension further elongated the support set of the shortest vector, and the hyperparameters originally applicable at 40D faced the risk of becoming ineffective.

3.5.1 Comparative Experiment 1: The Ultimate Architecture Duel (MLP vs Set Transformer)

Targeting the complex inter-row interactions at 50D, we hypothesized that the traditional Multi-Layer Perceptron (MLP) might not capture long-range dependencies,

so we designed a V50-ST (Set Transformer) based on the self-attention mechanism. This network extracted each row into 100-dimensional features (its own length, cos with other rows, and GSO-μ with other rows) and input them into a 2-layer 4-head attention network.

- **Result**: V50-ST suffered a Waterloo, with a breakthrough rate of only **3.00%**, and a gap to BKZ45 as high as 63.77%.
- **Conclusion**: When the action space itself is already highly structured (continuous sliding window), the complex Attention mechanism not only brought no benefits but also led to severe overfitting and low sample efficiency due to its massive parameter count. In contrast, traditional MLP, with its strong inductive bias, stably extracted effective policies within 300,000 steps.

3.5.2 Comparative Experiment 2: Support Set Coverage Law and the Leap of K Value (K=35 vs K=40)

This was the most exciting discovery in the entire research. In 50D, the non-zero coefficient support set of the shortest vector typically spans between 25 and 40 rows.

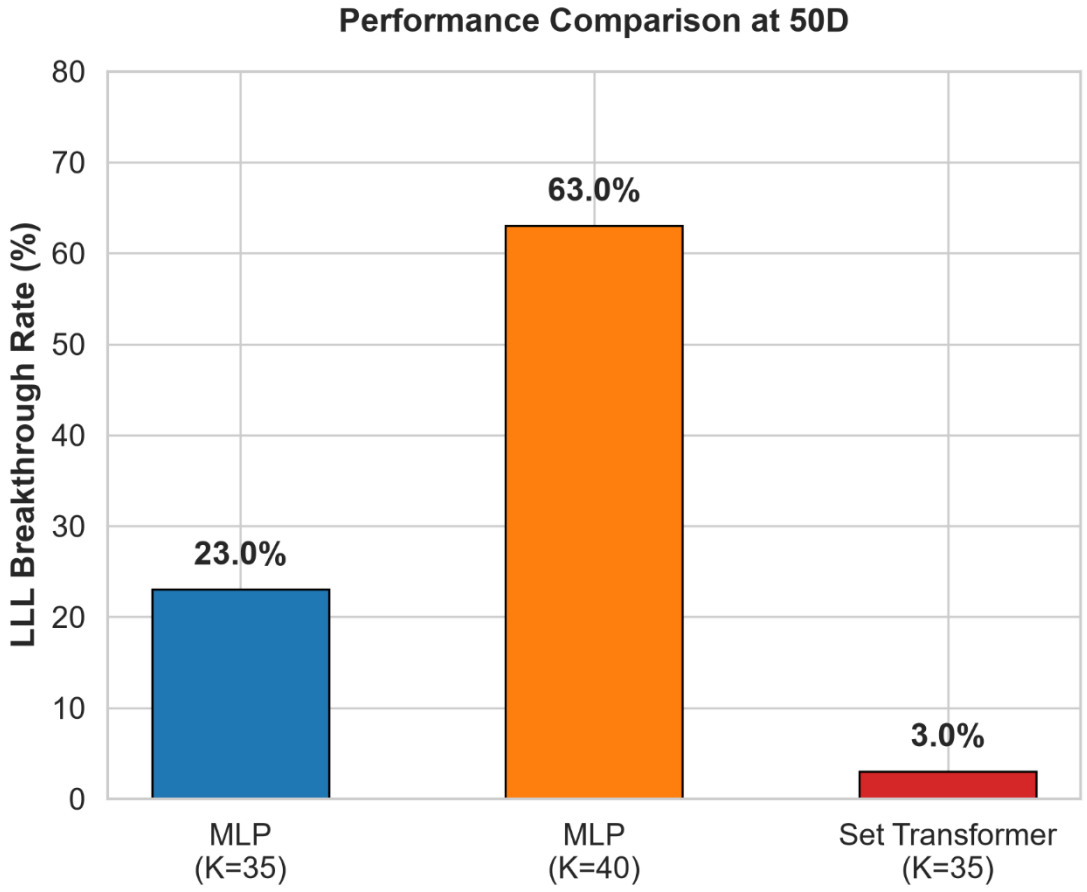


Figure 2: Performance Comparison at 50D

As shown in Figure 2, the comparison results reveal a dramatic difference:

- **V50-CONTIG.2 (K=35)**: At that time, the sliding window size was set to 35. Although the model showed effectiveness, the breakthrough rate stalled at **23.0%**. This is because, for some extreme lattices, the support set exceeded 35 rows, and the window failed to cover it perfectly.
- **V50-CONTIG.3 (K=40)**: We decisively increased the sliding window size to 40 (accounting for 80% of the dimension). This parameter change triggered a miracle:
    1. The LLL breakthrough rate surged from 23% to **63.0%**!
    2. The average gap to LLL dropped to **-1.21%** (i.e., the average vector length found by the AI was even shorter than standard LLL(0.99)).
    3. The average gap to BKZ(45) was drastically reduced from 50.47% to **18.61%**.
    4. The worst-case single performance improved by nearly 50% (from 11.91 million to 7.85 million).
- **Conclusion**: At 50D, "K=40" is the golden key to unlocking the treasure trove of extreme lattice performance. When the window size exactly covers the support set limit of high-dimensional short vectors, the power of sublattice BKZ enumeration is 100% unleashed. Simply and brutally expanding the effective field of view yields far greater benefits than complex network architecture optimization.

3.6 Chapter Summary

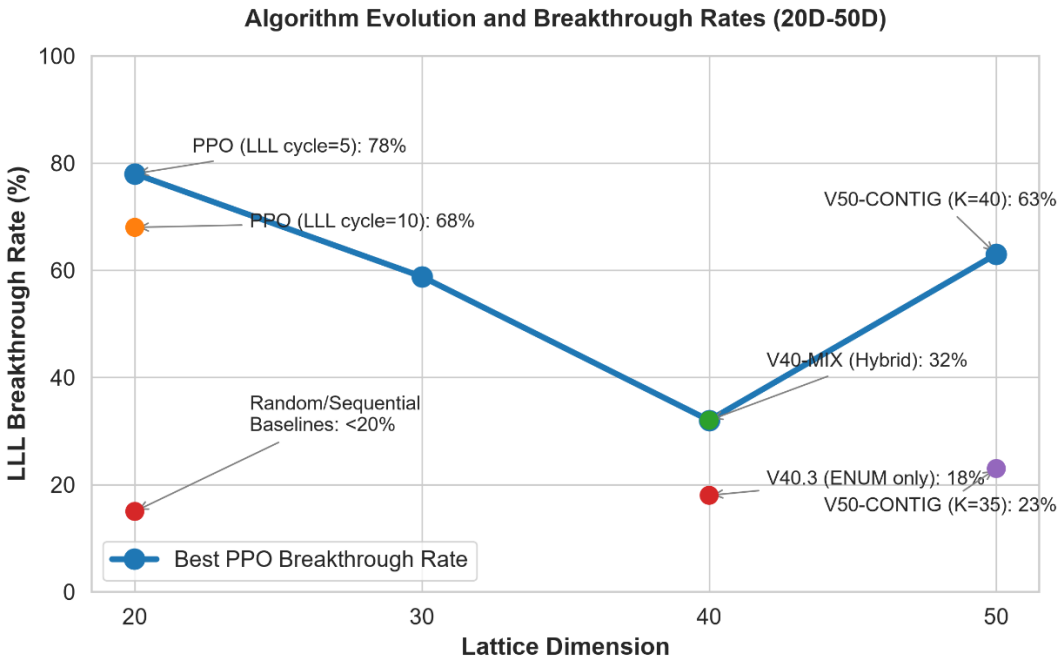


Figure 3: Algorithm Evolution and Breakthrough Rates (20D-50D)

As shown in Figure 3, the evolution from 20D to 50D is a scientific journey from "local fine-tuning" to "global search," and from "blind trial and error" to "following mathematical intuition." We have proven in high-dimensional lattice cryptography problems: local 2D/3D actions have mathematical deadlocks; complex high-dimensional feature concatenation causes neural network signal collapse; and the sublattice enumeration strategy based on local continuity, combined with a window size sufficient to cover the true support set (such as K=40 at 50D), is currently the optimal paradigm for Reinforcement Learning to conquer the SVP problem. The ultimate 63% breakthrough rate achieved at 50D provides a highly promising heuristic acceleration scheme for post-quantum cryptanalysis.

4. Engineering Optimization and Underlying Architecture

To support the aforementioned high-dimensional lattice cryptography experiments from 20D to 50D and overcome the computational disaster caused by 10901090-level ultra-large integers, the system underwent deep reconstruction and optimization in its underlying engineering architecture, which constitutes an indispensable cornerstone of this project:

1. Multi-process Parallel Computing Breaking Python GIL: In early versions, we used ThreadPool for environment sampling. However, sublattice BKZ enumeration is an

extremely CPU-intensive operation, and limited by Python's Global Interpreter Lock (GIL), multi-threading cannot achieve true parallelism. We switched the core environment to a multi-process architecture based on multiprocessing.Pool. 8 independent environments sample in parallel, and the time-consuming sublattice enumeration tasks are distributed to all CPU cores. This change increased the system's training throughput by dozens of times, making it possible to complete 300,000 steps of training in an extremely time-consuming action space like 50D.

2. Anti-Cheating and Numerical Shield in Extreme Large Integer Environments: In high-dimensional spaces, matrix determinants easily exceed 1030010300, and directly using floating-point numbers leads to severe overflow and physically non-existent "pseudo-zero vectors." The system introduced exact rational numbers (Fraction) and an integer backward verification mechanism based on sympy at the underlying level. When the agent claims to have found an extremely short vector, the system bypasses the neural network and directly infers its integer coefficients in the original lattice. This "anti-cheating" mechanism successfully captured a "monster pseudo-vector" reaching 9×10179×1017 generated by early code bugs, ensuring a 100% verification true rate.

3. Macro-Micro Synergistic LLL Fallback Mechanism: Comparative experiments proved (V40-NOL) that if completely relying on the AI's sublattice actions, the agent will fall into an exploration dead end above 40D, with the breakthrough rate dropping to 0%. Therefore, we designed a periodic LLL fallback mechanism (e.g., executed every 50 steps). The LLL algorithm plays the role of a "macro stabilizer," responsible for eliminating local chaos caused by AI exploration and continuously providing the agent with mathematically healthy "anchors"; while the AI acts as a "micro scalpel," conducting deep sublattice combination exploration on the basis of LLL. The organic combination of the two is the guarantee of convergence.

5. Conclusion, Core Innovations, and Future Work

Through extremely rigorous evolution, ablation, and comparative experiments from 20D to 50D, this paper not only successfully conquered some barriers of high-dimensional SVP problems but also precipitated a systematic methodology for AI to solve complex mathematical problems.

5.1 Summary of Core Innovations

Looking back at the entire research journey, we achieved three key innovations:

- **Paradigm Reconstruction (From Local to Global)**: At 20D-30D, we proved that 2D/3D Gaussian reduction has a mathematical deadlock after strong LLL

reduction. At 40D, we decisively upgraded the action space to "AI row selection + sublattice enumeration," completely opening up the high-dimensional search landscape.

- **Discovery of the Support Set Coverage Law**: In the 40D comparative experiment (V40.3 vs V40-CONTIG), we proved that the support set of short vectors after LLL has "local continuity." In the 50D experiment, this law was pushed to the extreme—we found that **the window size K is a hyperparameter that determines life or death**. When we increased K from 35 to 40, exactly covering the support limit of extreme lattice short vectors, it directly caused the breakthrough rate to undergo a qualitative change, skyrocketing from 23% to **63%**.
- **"Counter-intuitive" Validation of Architecture Choice**: At 50D, we tried the Set Transformer architecture (V50-ST), representing the cutting edge, attempting to capture inter-row relationships with self-attention. But the result (only 3% breakthrough rate) ruthlessly proved: under the premise that the action space is already highly structured (continuous sliding window), the complex Attention mechanism is not as effective as the strong inductive bias of traditional MLP. **Simple and precise features will always prevail over redundant and complex models.**

5.2 Future Work Outlook: Breaking the Computational Limit of a Single Agent

Although we achieved an astonishing 63% breakthrough rate at 50D, when the dimension continues to climb to 70D/80D, we will face an insurmountable physical ceiling. In 70D space, to cover the support set of the shortest vector, the window K must reach at least 55-60. However, the enumeration time for a single BKZ(60) will explode exponentially, and the computational power of a single agent will be unbearable.

Future research directions will inevitably turn to **Hierarchical/Recursive Reinforcement Learning Architecture** and **Sieving algorithms**:

1. **Divide-and-Conquer Strategy and Expert Invocation**: Drawing on the multi-level cache idea in computer systems. The top-level high-dimensional AI (e.g., 80D) no longer directly performs underlying enumeration but is responsible for global planning—it only decides "which regions are worth searching"; subsequently, it passes the selected sublattice (e.g., 50D) to a well-trained "50D expert model" with a high breakthrough rate for refinement; the 50D expert can even continue to invoke a 30D expert. Through this recursive invocation, the physical limit of a single agent is broken.

2. **Integration with Sieving**: When the dimension breaks through 80D, the enumeration method completely fails, and it is necessary to turn to the sieving algorithm (Lattice Sieving) with lower asymptotic complexity. Future AI will no longer learn "which rows to select for enumeration" but will learn "how to design optimal strategies for merging, collision, and elimination in a pool of hundreds of millions of candidate short vectors (sieving lists)."

This research not only provides a powerful AI solver for the SVP problem in lattice cryptography but also proves that Reinforcement Learning, when exploring complex combinatorial mathematical spaces, can discover local optimal paths that surpass human traditional algorithm designs.